VARIABLES IN SPATIAL LEARNING: II. DEGREE OF MOTIVATION DURING TRAINING AND TEST TRIALS[1]

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Inherent in any systematic approach to the field of learning is an acceptance of some criterion of learning. Not only do present day learning theories differ as to basic principles but also as to what constitutes an adequate test of the learning process.

Theories, based on principles derived largely from the data of conditioning have concerned themselves mainly with attempts at explanation of the formation and strengthening of simple associations. Attention is focussed upon relatively simple, easily measurable aspects of an organism's response (e.g. amplitude, latency) as it is modified during successive stimulations by a more or less unchanging environment. Learning is evidenced when the 'proper' modification in the response occurs (e.g. greater frequency of response in successive blocks of trials, reduction in latency, etc.).

Other theories, which for convenience of communication may be labelled 'field theories', have differentiated between the type of learning with which the conditioning theorist concerns himself and that which involves 'apprehension of relations'. Since the mere repetition or reproduction of a response is not sufficient evidence that learning or 'structuring' has occurred, it is frequently necessary to discover how the subject will react in a situation which is different but still contains certain of the properties of the original problem. If the second problem is solved the inference is made that the subject had learned the 'meaning' of the initial situation. At the descriptive level, this latter type of learning (apprehension of relations or structuring) reduces itself to the appearance of a new appropriate response in a novel problem solving situation.

We find therefore that for the conditioning theorist the reproduction of a response with certain modifications of its attributes is evidence of learning while to the field theorist such a criterion is not sufficient to infer that learning of the meaningful variety has occurred.

Although a selection of any criterion is in the final sense a function of individual taste and interests, it is quite obvious that the ultimate aim of all theories is to be able to interpret systematically the phenomena involved in both criteria of learning. Whether a system of postulates based on conditioning principles willl be adequate to predict the learning of the transfer variety, as well as of the response reproduction type, appears to be a question that only the future and the ingenuity of the conditioning theorist can answer. Several successful attempts appear to have been made (4, 8, 9). In spite of this the field theorists are obviously pessimistic about the success of such attempts by conditioning theorists.

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In the field of learning theories, Tolman has occupied a unique position, serving as a bridge between association theories and field theories. On the methodological level the pattern of theory construction of associationistic theorists, such as Hull (5), has been identical with the type initially espoused by Tolman (10). However, when it comes to the selection of specific principles and postulates in his behavior system, as well as areas of investigation, Tolman has obviously been stimulated by field theories. An expression of such stimulation has been the recent series of studies on spatial learning which have involved a transfer criterion of learning.

In the initial study (13) rats were originally trained to run down a path, involving several turns, to food. During the test trial the original path was blocked and the animals were offered a choice of 12 paths radiating from a round table top preceding the point of obstruction. Thirty-six per cent of the subjects chose the path which pointed directly towards the location of the goal. Such data were interpreted to indicate "that rats do learn to expect goals in specific locations". The authors felt that the data were inconsistent with stimulus-response formulations which "assert that what is learned in any spatial problem is a response tendency (i.e. a tendency to take the path on which the animal was trained). . . whenever the animal is appropriately motivated." Aside from the fact that Tolman and his associates give a very restricted meaning to the Hullian concept of 'response' it should be noted that in this study spatial learning was inferred when a large per cent of subjects were able to make a relatively new appropriate response to a novel situation. And the interpretation implies that it would be difficult, if not impossible, for the S-R theory to make the transition from the conditioning situation to the spatial learning problem.

Gentry, Brown, and Kaplan (3) repeating the work of Tolman, Ritchie, and Kalish (13), under essentially similar experimental conditions failed to confirm the results of the initial investigators. They found, with a much larger group of subjects, that the per cent of choices to the path which pointed towards the location of food was less than chance. This was true even when the amount of training on the original path was increased. A control group, having no experience in the training situation, behaved in a manner similar to the experimental group in the test situation. Without any additional information it appears impossible to reconcile the positive transfer effects found in the California study (13) with the contradictory evidence of zero transfer effects of the Texas study (3).

The present series of studies was undertaken after publication of the initial California study. The purpose was to investigate systematically the influence of certain variables, recognized as important in conditioning theory, upon spatial learning.

In the initial study, Kendler and Gasser (7) reported that varying the number of reinforcements had a differential effect upon spatial learning. The preliminary training in their experiment occurred in a free-choice non-correction T-maze situation. The population was divided into four groups, receiving respectively 0, 5, 20, and 100 reinforcements following their correct responses. During the twenty test trials the route to the choice point was blocked, and the animals had a choice of one of six possible pathways radiating symmetrically at angles of 31, 62, and 90 degrees from the new choice point. The results of the initial test trial provided no evidence of spatial learning but when the findings of the complete test series were analyzed the results indicated: (1) that varying number of reinforcements had a differential effect upon spatial learning, and (2) that spatial learning increased with greater number of reinforcements up to 20 but decreased at 100 reinforcements.

The present experiment was designed to investigate the influence of different degrees of motivation. Although most of the psychological theorists acknowledge the importance of motivation, its specific role in learning has not been adequately treated. According to Hull (5), whose theory is largely based upon conditioning principles, drive strength (motivation) is *not* one of the variables of which habit strength (learning) is a function. A study (6) in the field of drive interaction has justified this omission of drive strength as one of the determiners of habit

strength. It was found that animals, learning a bar depressing response under two different values of drive strength, had similar extinction records when the motivational conditions were equated during the extinction trials. Such data lend support to Hull's assumption that different degrees of motivation do not influence the amount of learning. The field theorists, preferring "a widely applicable, though quantitatively less precise system" (11) have not been as specific in their guesses as to the nature of the relationship existing between learning and motivation.

Tolman (12) has recently suggested that the cognitive map acquired under strong motivation may be narrowed. Translated into an experimental situation such an assumption would imply an inferior degree of spatial learning under strong motivation.

The methodology of this experiment was similar to that of the initial experiment of the present series. Two groups of animals, motivated by either 21 or 6 hours of food deprivation had similar experience during the training trials. The test trials were conducted under equated motivational conditions so that the influence of degree of motivation on the amount of spatial learning could be ascertained.

EXPERIMENTAL PROCEDURE

Subjects: The subjects for this experiment were 46 male albino rats of the Wistar strain purchased from the General Biological Supply House in Chicago, Illinois. At the beginning of the experiment they were approximately 70 days old.

Apparatus: A detailed description of the apparatus is presented in an earlier article (7). It is sufficient to say here that the construction of the apparatus permitted the animals to be subjected to two separate maze problems. During the training trials, the removable partition, indicated by the broken lines in the semi-circular area was put in place making a typical T-maze problem. During the test trials this partition was removed, the door in the semi-circular area leading to the choice point of the T-maze was lowered, and the animals were permitted to choose one of the six radiating alleys. The only light in the experimental room was that provided by a 150 watt reflector-flood bulb placed six feet directly above the choice point of the T-maze.

Preliminary Training: The animals were assigned randomly to four groups. Each animal was handled daily for two weeks. One week before the beginning of the experiment, the animals were placed on a 24 hour wet-food maintenance schedule that corresponded with the schedule used during the experiment. Special feeding cages were utilized and the rats were allowed to remain in these cages for 1½ hours during which they were permitted to consume a wet mash of Purina Dog Chow kibbled meal. The mean consumption was approximately 10 grams of food, not including water content, per rat per day. Following this feeding period, the animals were returned to their home cages, where water was always available.

The position habit of each animal was determined on the basis of three non-reinforced trials on the T-maze part of the apparatus.

Training Series: The four groups, which had been randomly selected, were labelled 6-6, 6-21, 21-6, and 21-21. The initial two groups contained 11 rats each while the latter two groups consisted of 12 animals each. The first number of the numerical identification refers to the number of hours of food deprivation prevailing prior to the daily experimental runs during the training series. The number following the hyphen refers to the motivation, based upon food deprivation, existing during the test trials. Thus two groups, 6-6 and 21-21 had similar motivational conditions during the training and test trials while the other two groups 6-21 and 21-6 had their motivation changed at the beginning of the test series.

The T-maze section of the apparatus was used during the training series. Approximately half of the animals were rewarded during the experiment on the side consistent with their position habit, while the remainder received their reward on their non-preferred side.

Each subject was given four free choice trials per day, using a non-correction technique, with at least 15 minutes elapsing between successive trials. If the correct alley was chosen the rat received $\frac{1}{16}$ teaspoonful of a fine mash (Purina Dog Chow checkers which had dissolved in water) in the goal box. This food was placed on an inverted metal paint-mixing dish whose diameter was $1\frac{1}{8}$ inches. If the incorrect alley was chosen the subject was confined in the empty goal box for a period of 30 seconds.

The training trials were continued until the subject had received 20 reinforcements, i.e. a total of twenty responses to the correct side followed in each case by the consumption of the goal object. Two animals, one in group 6-6 and the other in group 21-21 had to be subjected to a few forced trials to break their position habit to the incorrect side. The forced trials were counted as reinforcements.

Test Series: The day after the 20th reinforcement, the animal was placed in the test situation. The partition was removed from the semi-circular choice point and the door in the stem of the T-maze was lowered, thereby forcing the subject to choose one of the six

TABLE 1

Showing the number and per cent of each group responding to each of the six alleys on the first test trial

GROUP	N	CORRECT ALLEY											
		Alley 1		Alley 2		Alley 3		Alley 4		Alley 5		Alley 6	
		N	%	N	%	N	%	N	%	N	%	N	%
6-6	11	5	45.5	0	0.0	1	9.1	0	0.0	0	0.0	5	45.5
6-21	11	4	36.4	1	9.1	1	9.1	1	9.1	0	0.0	4	36.4
21-6	12	2	16.7	1	8.3	1	8.3	1	8.3	2	16.7	5	41.8
21-21	12	5	41.7	1	8.3	1	8.3	2	16.7	0	0.0	3	25.0
Total.....	46	16	34.8	3	6.5	4	8.7	4	8.7	2	4.4	17	37.0

radiating paths from the new choice point. Before the first trial of each rat, the semicircular choice compartment and a distance of one foot into each alley were wiped with a damp cloth to eliminate any possible odor which might have led to tracking. As in the training trials, each rat was given four trials per day. The amount and the location of the reward was the same as during the training trials except that the food was placed on an inverted clear glass coaster (diameter $1\frac{1}{2}$ inches).

In the recording system used to identify the six alternative choices during the test series, alley 2 represented the choice of the correct alley. If an animal, during the training trials, received food in the goal box (GB1) on the left side of the maze, then the alleys were numbered from 1-6 from left to right, with alley number 2 leading to the baited goal box. If on the other hand, the animal found food in the right goal box (GB2), the numbering system was reversed by numbering the alleys from 1-6 from right to left. With the use of such a numbering system alley 2 always designated the path leading directly to the baited end-box.

At the beginning of the experiment it was planned to have each rat reach a criterion of 12 successive responses to alley 2. At the end of 10 days of running during the test trials only a few rats had reached the criterion, while most of the others indicated no appreciable signs of learning. It was decided to use a 20 test trial series to evaluate the influence motivation has upon spatial learning. Such a criterion would be identical with the one used in the initial study (7).

RESULTS

The results of the test trials are analyzed in terms of the performance exhibited by the various groups on the first test trial as well as the entire test series.

Table 1 presents the results of the first test trial and it indicates no appreciable difference between the performance of the four experimental groups. Only three of the 46 animals (6.52 per cent) responded to the test situation by choosing alley 2 while the majority of the animals (71.74 per cent) chose either alley 1 or alley 6.

TABLE 2

Showing the mean number of responses exhibited by each group to each of the six alleys during the 20 test trials

GROUP	CORRECT ALLEY											
	Alley 1		Alley 2		Alley 3		Alley 4		Alley 5		Alley 6	
	M	σ	M	σ	M	σ	M	σ	M	σ	M	σ
6-6	7.09	5.05	4.18	5.13	1.00	1.21	.73	.98	1.46	1.16	5.55	5.07
6-21	5.55	3.45	3.18	2.17	2.27	2.26	1.09	1.24	1.27	1.29	6.64	5.09
21-6	3.83	2.48	4.83	4.24	2.25	1.83	1.42	1.18	1.75	1.30	5.92	4.75
21-21	6.33	4.59	5.08	3.35	1.92	1.04	1.17	.99	1.92	1.26	3.58	2.06

TABLE 3

Comparison of the mean number of correct responses during the test series between groups that had different motivations during the training series but the same motivation during the test series

COMPARISON	MEANS	D.F.*	F	LEVEL OF SIGNIFICANCE
21-6/6-6	4.83/4.18	31/22	1.16	—
21-21/6-21	5.08/3.18	55/47	1.60	20%

* The degrees of freedom for two groups (5 and 20) were incorrectly reported in the first paper of this series. However, this mistake did not appreciably affect the interpretation of the F-ratio.

Table 2 shows the mean number of responses exhibited by each group to each alley during the 20 test trials. Although the response to alley 2 was the only one rewarded during the test series, the subjects in each group chose the two 90 degree alleys (alleys 1 and 6) more frequently than the rewarded path.

In order to discover the role of drive in spatial learning an analysis of variance type of statistical treatment of the correct responses for the various groups would have been particularly appropriate for this type of experimental design. However this type of statistical treatment was precluded because of the unequal variances and skewed distributions of the obtained data. A technique devised by Festinger (2) which is appropriate for testing the reliability of means from

skewed distributions was used to obtain the significances of the differences shown in tables 3 and 4.

Table 3 presents the comparison between those groups with different motivation during the training series and the same motivation during the test series. One of these comparisons was significant at the 20 per cent level of confidence. This is obviously not sufficient to reject confidently the null hypothesis, namely that the amount of spatial learning is independent of the degree of motivation during the training trials.

Table 4 presents the comparison between those groups with the same motivation during the training series and different motivations during the test series. Neither of these differences approached significance and consequently the null hypothesis, namely that the amount of spatial learning evidenced is independent of the degree of motivation during the test trials, cannot be rejected.

TABLE 4

Comparison of the mean number of correct responses during the test series between the groups that had different motivations during the test series but the same motivation during the training series

COMPARISON	MEANS	D.F.	F	LEVEL OF SIGNIFICANCE
6-6/6-21	4.18/3.18	22/22	1.31	—
21-21/21-6	5.08/4.83	55/31	1.05	—

DISCUSSION OF RESULTS

The results of the first test trial are similar to those obtained in the initia study of the series (7). In that study, in which number of reinforcements during the training series was the variable investigated, 5.6 per cent of the animals exhibited a preference for alley 2 (the alley leading directly to the location of the food) as compared with 6.52 per cent of the animals of the present experiment. However, the animals of the initial study exhibited a preference for alley 1 over alley 6. This was due to the fact that the relative preference for alley 1 over alley 6 was related to the number of reinforcements. The group of animals who received 100 reinforcements during the training series had a decided preference for alley 1.

The results of these two investigations provide no evidence that the animals on the first test trial select the path leading directly toward the location of the goal. In fact the choice of this path (alley 2) was less than chance. However, there is some indication that the response on the first test trial was to some extent determined by emotional factors. It was found in an experiment (1) in which the light source was above the baited goal box during the training series that the animals favored alley 6 on the first trial. In such an experimental arrangement, alley 6 was furthest away from the light and consequently was the darkest alley. In the present study the light was directly above the T-maze choice point. This resulted in alleys 1 and 6 being the most shaded. It is felt that such evidence implies that an important variable in our studies during the initial test trial (as

well as in the entire test series) was the brightness of the various alternative alleys. It appears that a dim light source would have been more effective in ascertaining the basic variables of spatial learning.

The relatively few choices of alley 2 during the 20 trial test series should not be interpreted as evidence of zero transfer effects between the training and test series. The small number of such responses probably reflects, to some extent, the brightness variations between the different alleys. However it should be remembered that in the initial study (7) a group having no training on the T-maze made a mean of 2.50 correct responses during the 20 test trials. The data obtained from the present experimental groups and the remaining groups of the initial study indicate the existence of some slight positive transfer effects between the training and test trials.

The data relating most directly to the problem of the relation between the degree of motivation and the amount of spatial learning were obtained by comparing the number of correct responses made by the different groups during the test series. The initial comparison (table 3) revealed that the intensity of motivation during the training series had no effect upon the amount of spatial learning revealed during the test series.[2] Such a finding would be consistent with Hull's assumption that drive strength in *not* one of the variables of which habit strength is a function. At least in this instance the postulate concerning the relationship between motivation and learning has been able to make the transition from the simple conditioning situation to the somewhat more complicated spatial learning problem.

For Tolman the evidence would indicate that the cognitive maps acquired by these rats under either 6 or 21 hours of food deprivation were equally 'broad' (12). Obviously these results are not inconsistent with his assumption that too strong motivations might narrow the cognitive map. It would be interesting to observe whether a *hunger drive* based upon a much greater deprivation period than 21 hours, would yield an inferior amount of spatial learning in an experimental situation similar to ours.

The finding that the degree of motivation during the 20 test trials had no effect upon the performance of the animal during the test trial appears somewhat surprising. According to Hull, drive strength acts upon habit strength in a multiplicative manner. In a spatial learning situation many S-R associations are being formed, and a stronger drive although increasing the excitatory potential of the individual S-R associations would not effect their rank order of

[2] One might argue that the slight superiority exhibited by the groups having a 21 hour drive during the training series should not be overlooked. Such a suggstion should be considered but it should be noted that such a slight superiority need not be ascribed to the facilitating effect upon learning a 21 hour hunger drive has as compared to a 6 hour hunger drive. The habit strength to go towards the cues of the baited goal box was probably slightly greater for the former group. This would be due to the fact that the 21 hour hunger group exhibited significantly faster running time. Consequently the time interval occurring between making the correct response and the consumption of the goal object during the training trials would be less for the 21 hour group. Delay of reinforcement is one of the variables inversely related to habit strength (5).

strengths. One would however suppose that the animals with the stronger motivation during the test trials should more rapidly achieve a greater proficiency in the selection of alley 2. However because of the slight degree of positive transfer to the test situation and the apparent difficulty of the test problem it may be necessary to have a much longer test series in order to demonstrate the superiority of the more highly motivated groups. The animals which failed to achieve a criterion of 12 successive responses to alley 2 were subjected to the test situation until they had completed 40 trials. Assuming that those subjects which achieved the criterion would continue to choose alley 2, it was found that at the end of the 40 trials, the groups with the 21 hour hunger drive during the test trials had slightly more, although not significantly more, choices of alley 2 than did the groups motivated by 6 hours of food deprivation.

SUMMARY

The purpose of this study was to investigate the relationship of motivation to the degree of spatial learning. Forty-six animals were divided into four groups having either 6 or 21 hours of food deprivation. In two groups, the motivation was changed between the training trials and the test trials, and in the other two groups the motivation was held constant during the experiment.

An apparatus was employed which made possible both the use of a T-maze during the training series and a maze with six choices in the test series, one of which led to the baited goal box of the T-maze. A light was placed directly over the choice point of the T-maze. In the training series the animals received 20 reinforcements on the T-maze following the correct choice. After this training, 20 test trials were given on the portion of the maze which involved the six alternative paths.

On the initial test trial the choice of the path leading to the baited goal box was less than chance. There was no significant differences between the performances of the different motivational groups. It was also found that motivation did not affect the degree of spatial learning as measured by the number of correct responses during the 20 trials of the test series. The theoretical implications of these results are discussed.

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